

WASP-43b

R-0472

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_231225 · created 2026-07-20T01:55:44+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460303.9621 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.02
Transit depth (Rp/Rs)^2	2.42 +/- 0.61 [%]
Orbital Inclination (inc)	80.68 +/- 0.99
Airmass coefficient 1 (a1)	1.043 +/- 0.01
Airmass coefficient 2 (a2)	-0.03 +/- 0.025
Scatter in the residuals of the lightcurve fit is	0.97 %
Best Comparison Star	None
Optimal Aperture	3.92
Optimal Annulus	15.59
Transit Duration (day)	0.0456 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.155 $\pm$ 0.02 vs 0.1594 (deviation 0.22 σ)
Duration fit vs published	0.0456 d vs 0.0512 d (deviation 1.26 σ)
Mid-time O–C	0.9 min
Depth SNR	7.8
Residual scatter	0.97 %

Light curve

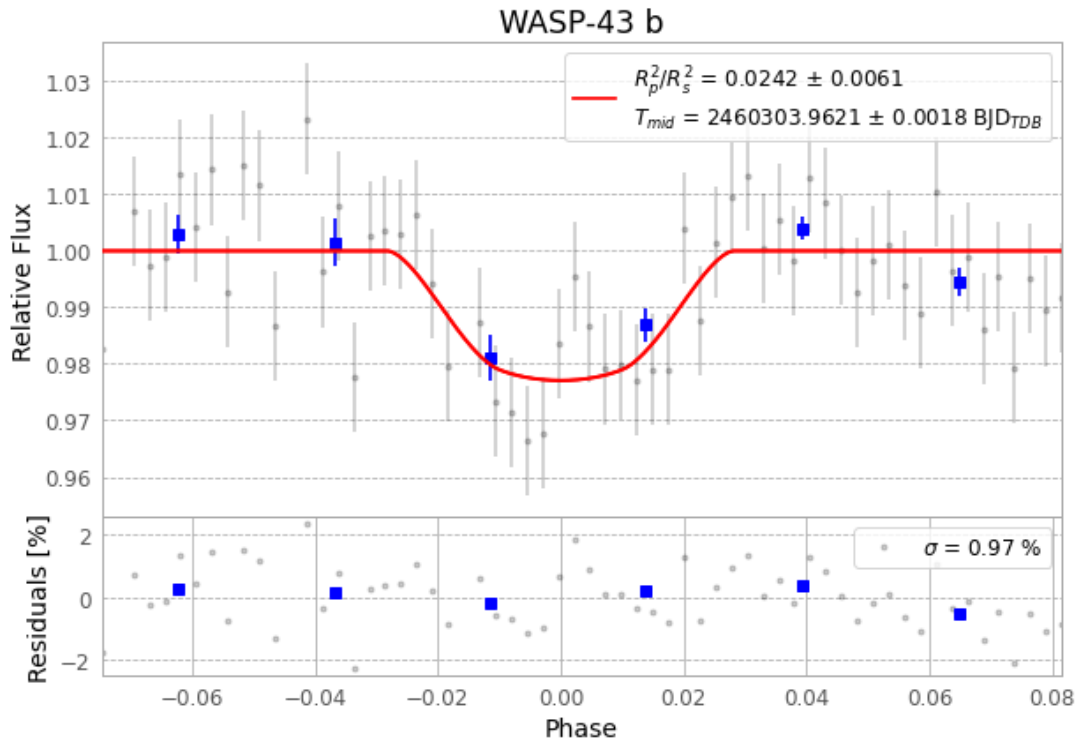


Figure 1 — FinalLightCurve_WASP-43 b_2023-12-25.png (SHA-256 61dbc2cdd2c891f7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [473, 259], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dd7a1cb49c7d641b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

62 FITS frames (41 MB), WASP-43231225093315.FITS → WASP-43231225123615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-231225.log (10d24dd27a71...), FinalLightCurve_WASP-43 b_2023-12-25.png (61dbc2cdd2c8...), FinalLightCurve_WASP-43 b_2023-12-25.csv (6dcdd0100392...)

Compute resources

Wall time 105.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 01:55Z.